

AIRFLOW INSTALLATION

1. Install Airflow (Exact Version)

```
# create project folder
mkdir airflow_stacd && cd airflow_stacd

# create venv
python3 -m venv venv
source venv/bin/activate

# upgrade pip
pip install --upgrade pip

# set airflow home (IMPORTANT)
export AIRFLOW_HOME=$(pwd)/airflow
mkdir -p $AIRFLOW_HOME

# install airflow 2.10.4 with constraints
AIRFLOW_VERSION=2.10.4
PYTHON_VERSION=$(python -c "import sys; print(f'{sys.version_info.major}.{sys.version_info.minor}.{sys.version_info.micro}')")

pip install "apache-airflow==${AIRFLOW_VERSION}" \
  --constraint "https://raw.githubusercontent.com/apache/airflow/constraints-${AIRFLOW_VERSION}"
```

2. Clone Your Repos

Make another folder or if you prefer inside the same airflow folder itself and clone the dev branch.

```
git clone -b dev https://github.com/SaharshLaud/STACD_framework
```

3. Setup Folder Structure

We want Airflow to see your enhancement folder exactly like the current system.

```
# copy enhancement folder  
cp -r STACD-Airflow/enhancement $AIRFLOW_HOME/  
  
# copy plugins folder  
cp -r STACD-Airflow/plugins $AIRFLOW_HOME/
```

4. Set PYTHONPATH

```
export AIRFLOW_HOME=$(pwd)/airflow  
export PYTHONPATH=$AIRFLOW_HOME:$AIRFLOW_HOME/enhancement:$PYTHONPATH
```

(Optional permanent)

```
echo "export AIRFLOW_HOME=$(pwd)/airflow" >> ~/.bashrc  
echo "export PYTHONPATH=\$AIRFLOW_HOME:\$AIRFLOW_HOME/enhancement:\$PYTHONPATH" >> ~/.bashrc
```

5. Initialize Airflow DB

```
airflow db init
```

Create user:

```
airflow users create \  
  --username admin \  
  --firstname admin \  
  --lastname admin \  
  --role Admin \  
  --email admin@example.com
```

6. Initialize STACD DB (YAML → DB)

Go to enhancement/database:

```
cd $AIRFLOW_HOME/enhancement/database
```

Run:

```
python init_db.py \  
--dag corestack_lite_dag.yaml \  
--algo-repo corestack_lite_algorithm_repo.yaml \  
--dataset-repo corestack_lite_dataset_repo.yaml
```

What this does:

- Registers Algorithm Instances
- Registers Algorithm Types
- Registers Dataset Types
- Registers Root datasets
- Registers DAG
- Generates STAC-D items

7. Generate DAG from YAML

Go:

```
cd $AIRFLOW_HOME/enhancement/dag_generator
```

Typical command (you may already have a wrapper, but generic way):

```
python stacd_recompute_generator.py \  
--dag-yaml ../yaml_configs/corestack_lite_dag.yaml \  
--output ../generated_dags/corestack_lite_dag_generated.py
```

Then copy DAG to airflow:

```
cp ../generated_dags/corestack_lite_dag_generated.py $AIRFLOW_HOME/dags/
```

8. Run Airflow

Standalone mode:

```
airflow standalone
```

OR

Production mode:

Start scheduler:

```
airflow scheduler
```

In another terminal:

```
source venv/bin/activate
export AIRFLOW_HOME=$(pwd)/airflow
export PYTHONPATH=$AIRFLOW_HOME:$AIRFLOW_HOME/enhancement:$PYTHONPATH

airflow webserver --port 8080
```

9. Trigger DAG

UI → <http://localhost:8080>
